ZYNQ_MINI Hardware Specification (English translation)

Translated from `02ZYNQ_MINI□□□□□□□□□□.pdf` *("□□□□□ ZYNQ □□□□" / Little Panda Academy ZYNQ teaching document; authors: Teacher Yang, Teacher Wu; finalised 2022-10-12). Board: Bochen Jingxin (□□□□) ZYNQ MINI, rev. B (silkscreen 20221015/REVB), XC7Z010-1CLG400.*

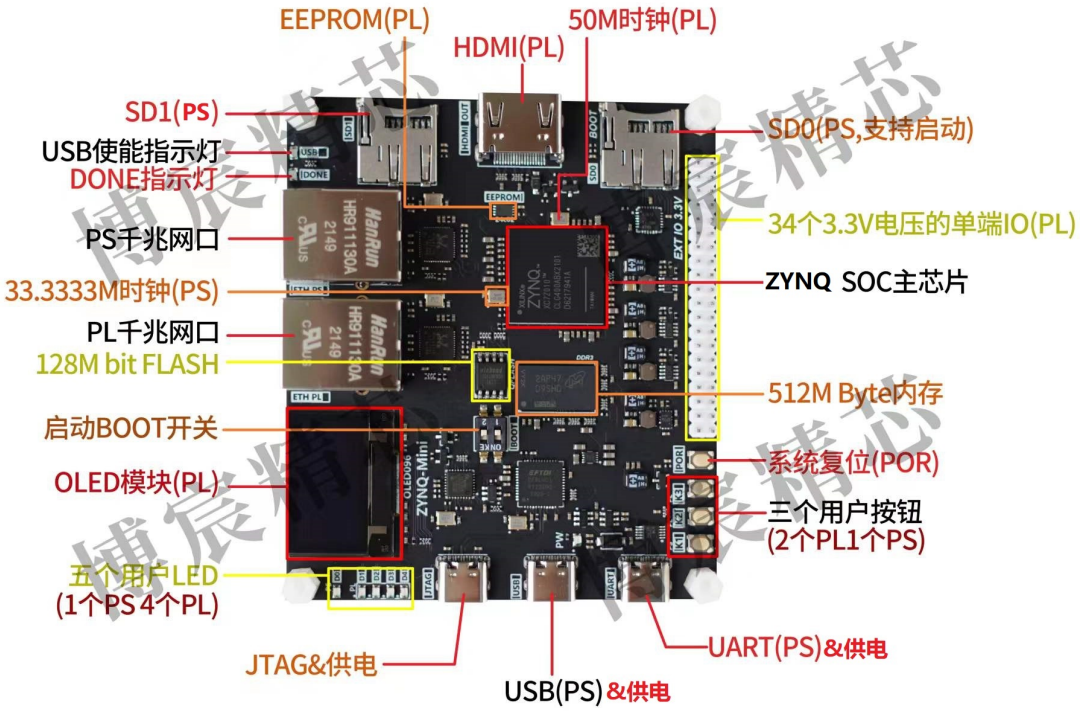
Cross-checked against `ZYNQ_MINI_REVB□□□.pdf` (schematic), the back-side silkscreen photo, the vendor example projects (Vivado 2018.3) and the vendor XDC files. Items that come from those sources rather than the PDF text are marked **[verified]** or **[added]**. The circuit figures of the original are re-cropped from the source PDF into each section (`figures/sec*.png`; their labels remain in the original language/schematic notation).

Contents: 1 Introduction · 2 Power · 3 Clocks and boot · 4 Debugger and UART · 5 DDR3 · 6 Gigabit Ethernet · 7 Storage · 8 Expansion IO, LEDs, buttons · 9 USB · 10 HDMI · 11 OLED · 12 Pin summary

1. Introduction

This document summarises the board's specification: functions, parameters, and an explanation of the circuit implementation. It targets the ZYNQ_MINI board; the older ZYNQ_ECO series is similar in function and hardware approach and can also use it as a reference.

In Zynq, **PS** is the ARM processing system and **PL** is the FPGA fabric; these abbreviations are used throughout.



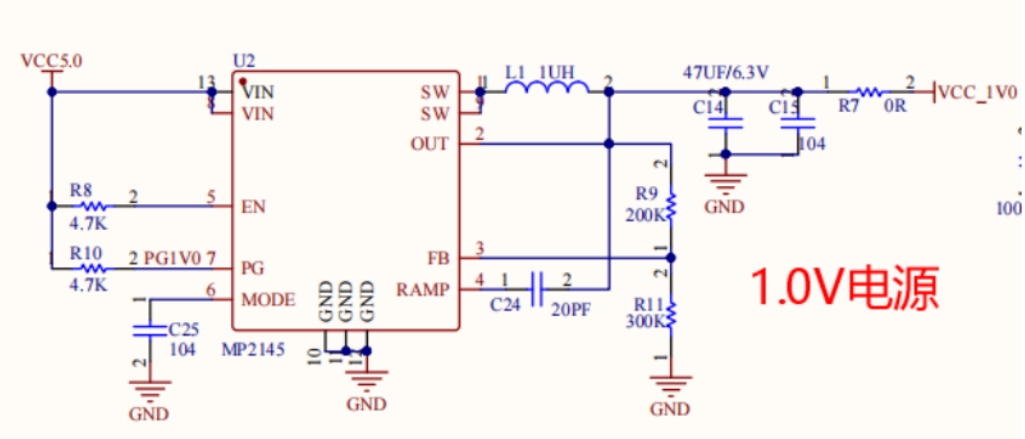
Interface labels in the photo (translated):

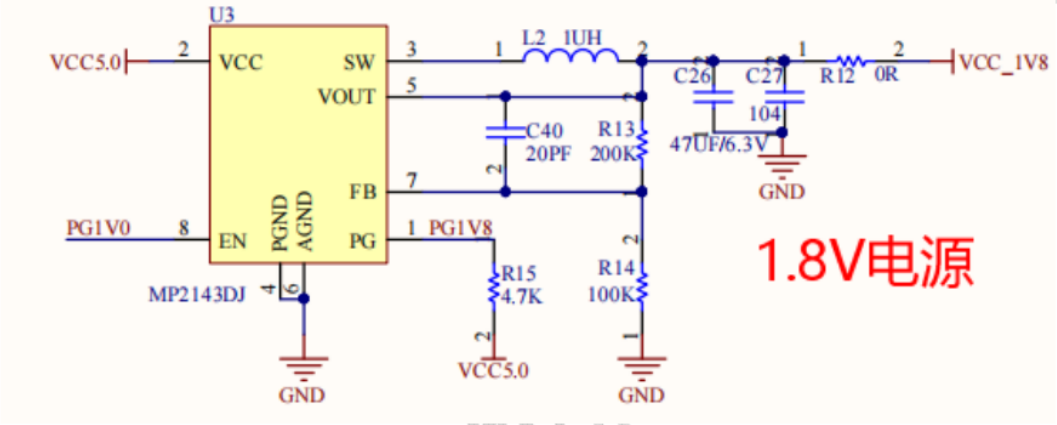
Label	Meaning
EEPROM (PL)	24C02 EEPROM on PL
HDMI (PL)	HDMI output on PL
50M clock (PL)	50 MHz PL oscillator
SD1 (PS)	TF slot 1, PS (MIO 10–15)
SD0 (PS, boot capable)	TF slot 0, PS (MIO 40–45)
USB enable LED / DONE LED	USB power-enable indicator; FPGA configuration-done indicator

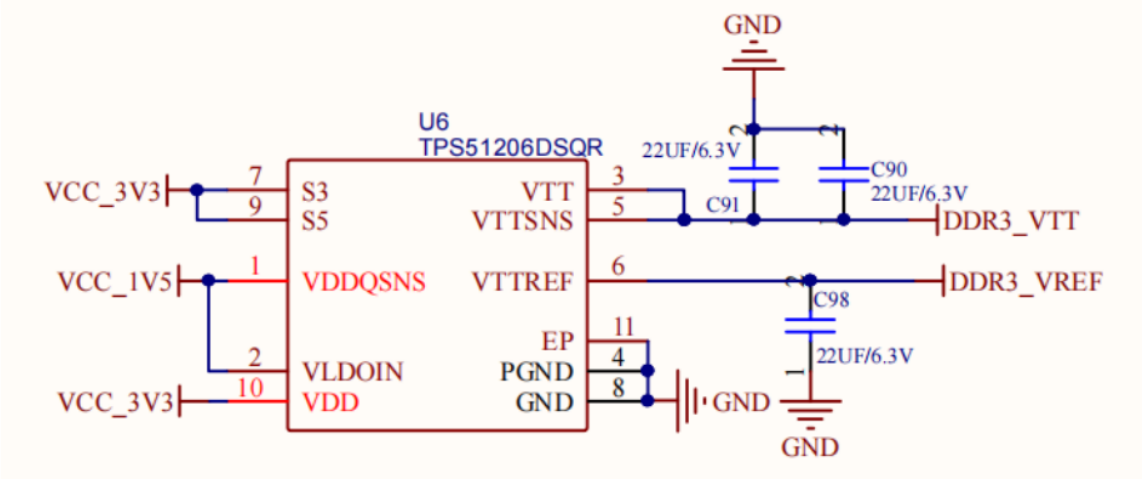
Label	Meaning
PS Gigabit port / PL Gigabit port	RTL8211E Ethernet on PS (MIO) / on PL (BANK35)
33.3333M clock (PS)	PS oscillator
128M bit FLASH	W25Q128FV QSPI flash
BOOT switch	SW1 boot-mode DIP switch
OLED module (PL)	0.96" SPI OLED
Five user LEDs (1 PS, 4 PL)	D6 (PS), D2–D5 (PL)
34 × 3.3 V single-ended IO (PL)	40-pin 2.54 mm header, BANK34
ZYNQ SoC	XC7Z010-1CLG400
512 MByte memory	DDR3
System reset (POR)	POR button
Three user buttons (2 PL, 1 PS)	K1 (PS), K2/K3 (PL)
JTAG & power / USB (PS) & power / UART (PS) & power	three Type-C connectors

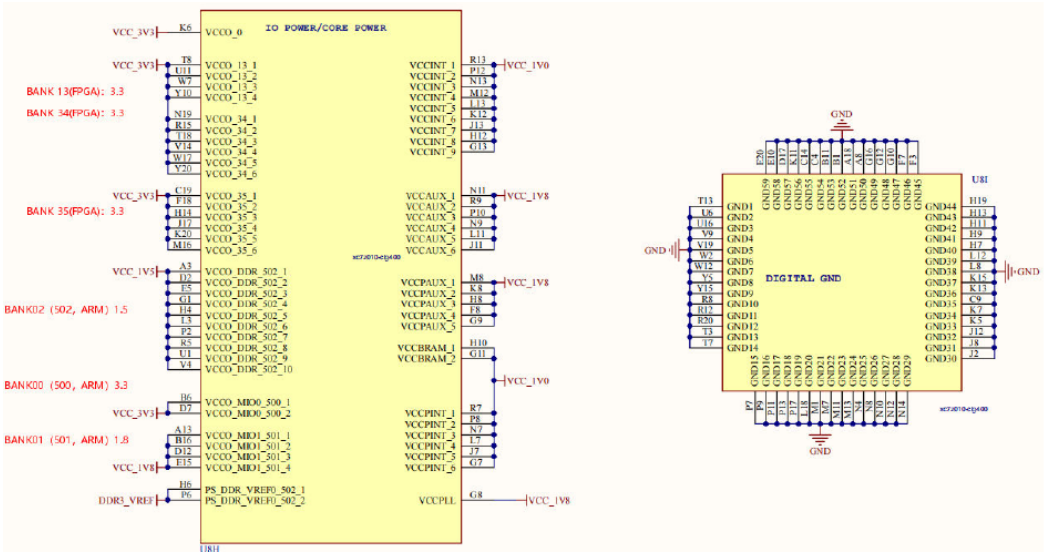
Note: JTAG, USB and UART connectors can each power the board during development. The USB connector can also supply power outward when operating as a USB host.

2. Power supply design









Schematic parts 1 and 2.

Five supply functions: **1.0 V** core (FPGA internal logic), **1.5 V** (DDR3 IO), **1.8 V** (VCCAUX, PLLs, 1.8 V IO), **3.3 V** (3.3 V IO), and the DDR3 termination/reference supply.

- **1.0 V core**: MP2145, synchronous-rectifier, up to **6 A**. It is separate from the other rails because the board family also ships with the 7020 (85K logic cells); with logic and clock rates pushed hard the estimated core current is 4–5 A (a 7010 needs only 2–3 A).
- **1.8 V, 1.5 V, 3.3 V**: MP2143DJ, 3 A synchronous converters (same circuit for each rail).
- **Sequencing**: Zynq requires **1.0 V → 1.8 V → 1.5 V → 3.3 V**. Each converter's PG output (pulled up to 5 V when the rail is in regulation) drives the next converter's EN. The 1.0 V converter's EN is tied high.
- **Power-on reset**: after the 3.3 V rail comes up, a **MAX809T** supervisor asserts reset to the PS **PS_POR_B** pin (BANK500, schematic part 4). When POR releases, the chip samples the boot mode and loads: QSPI boot reads flash; SD boot reads SD0. The same signal is available through the SYS RST button.
- **DDR3 reference**: TI **TPS51206** produces **VTT** and **VREF** (≈0.75 V, half of the 1.5 V input). VTT terminates the address/control lines (39 Ω pull-ups) and must source >2 A transient current; VREF is the IO reference. (Parts such as TPS51200 need two external equal resistors, e.g. 10 kΩ, to set the reference.)

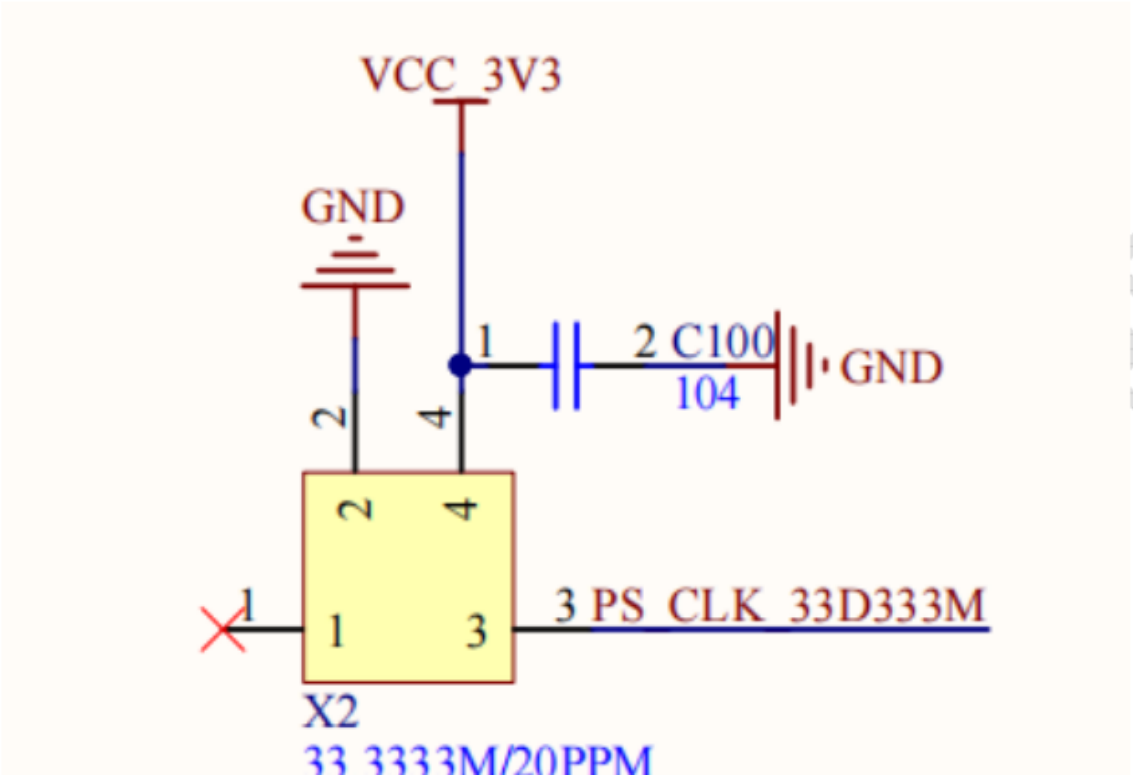
Bank supply voltages (the voltage in constraints must match):

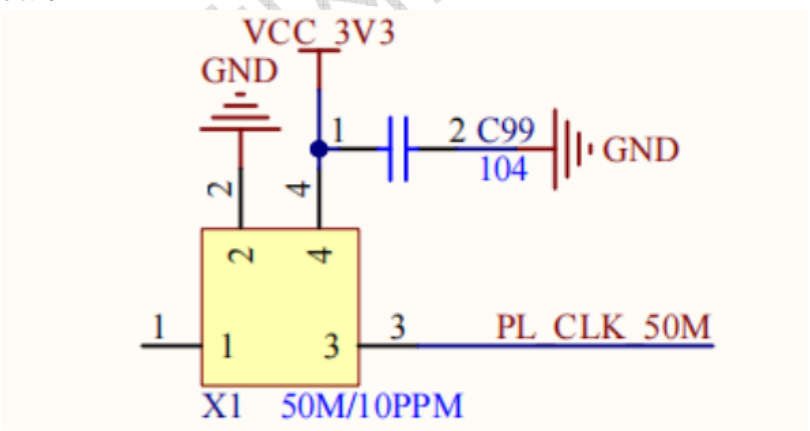
Bank	Domain	Voltage
13	PL (7020 only; not connected out on this board)	3.3 V
34	PL	3.3 V
35	PL	3.3 V
500	PS MIO 0–15	3.3 V
501	PS MIO 16–53	1.8 V

Bank	Domain	Voltage
502	PS DDR	1.5 V

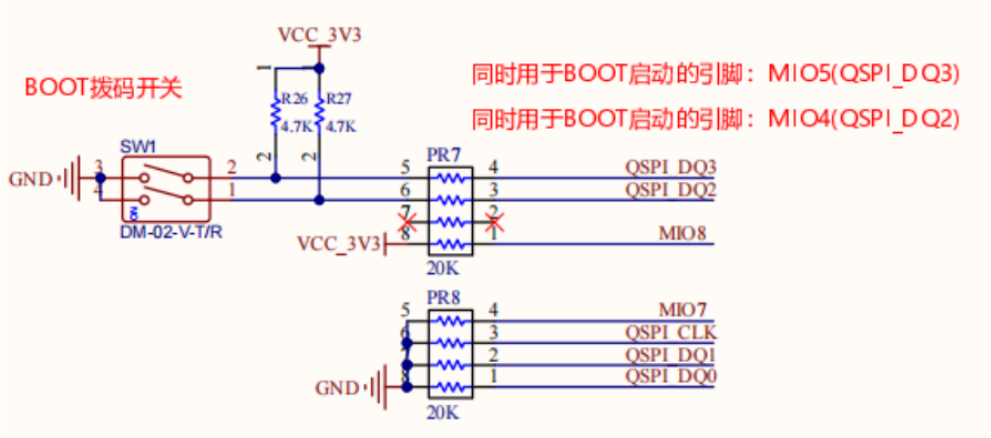
Internal rails: VCCPINT (PS core) 1.0 V; VCCINT (PL core) 1.0 V; VCCAUX (PL aux) 1.8 V; VCCPAUX (PS aux) 1.8 V; VCCBRAM 1.0 V; VCCPLL 1.8 V.

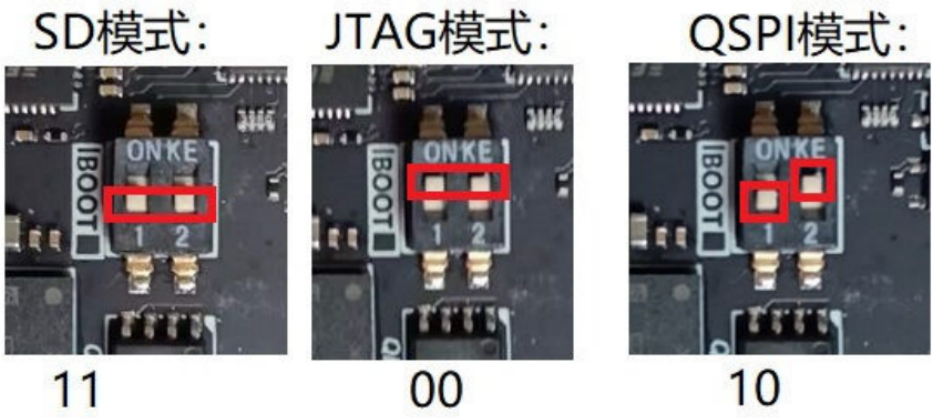
3. Clocks and boot circuit





PL时钟给FPGA部分提供时钟源，单端3.3V电压，连接到BA





Two oscillators (schematic parts 4 and 5):

- **PS clock**: 33.3333 MHz (PS_CLK, BANK500 pin E7). [verified]
- **PL clock**: **50 MHz**, single-ended 3.3 V, to BANK35 pin **K17** (`IO_L12P_T1_MRCC_35`). In Zynq-7000, PL clock inputs must use clock-capable pins: MRCC pins reach the global clock network, SRCC pins the regional network. A single-ended oscillator must go to the **P** pin of an SRCC/MRCC pair, never the N pin; differential clocks use both P and N.

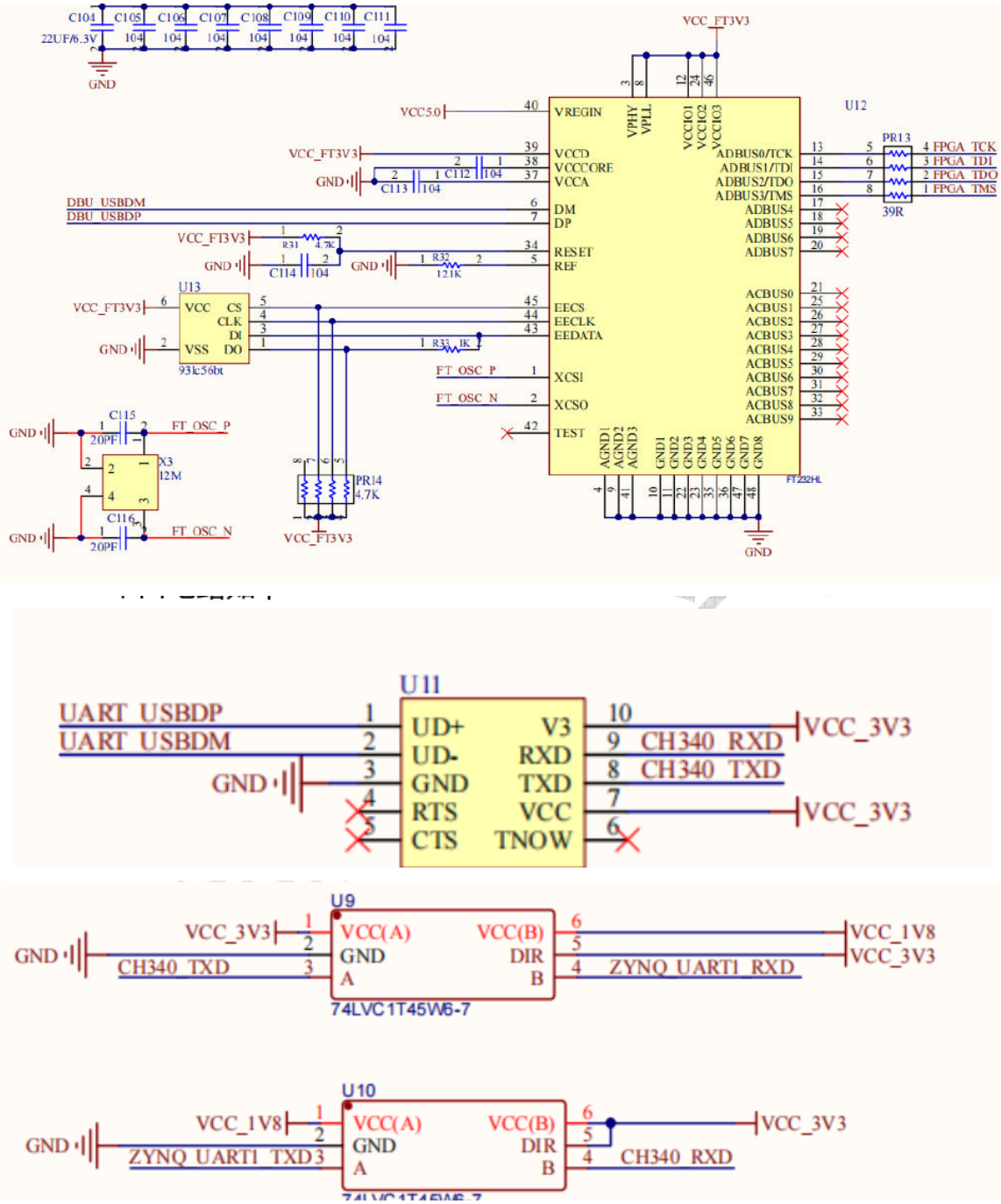
Boot mode (SW1, schematic part 4; silkscreen on the back of the board):

Mode	ON KE[1:2]
JTAG	00
QSPI	10
SD card (SD0)	11

The side of the switch marked "ON KE" is **0**; the side marked with the numbers 1, 2 is **1**. The boot-mode pins are shared with MIO4 (QSPI_DQ2) and MIO5 (QSPI_DQ3) (schematic part 4).

Note from the original: JTAG debugging works in any boot mode; in JTAG mode the XADC temperature can be read through the PC tool, in other modes it cannot.

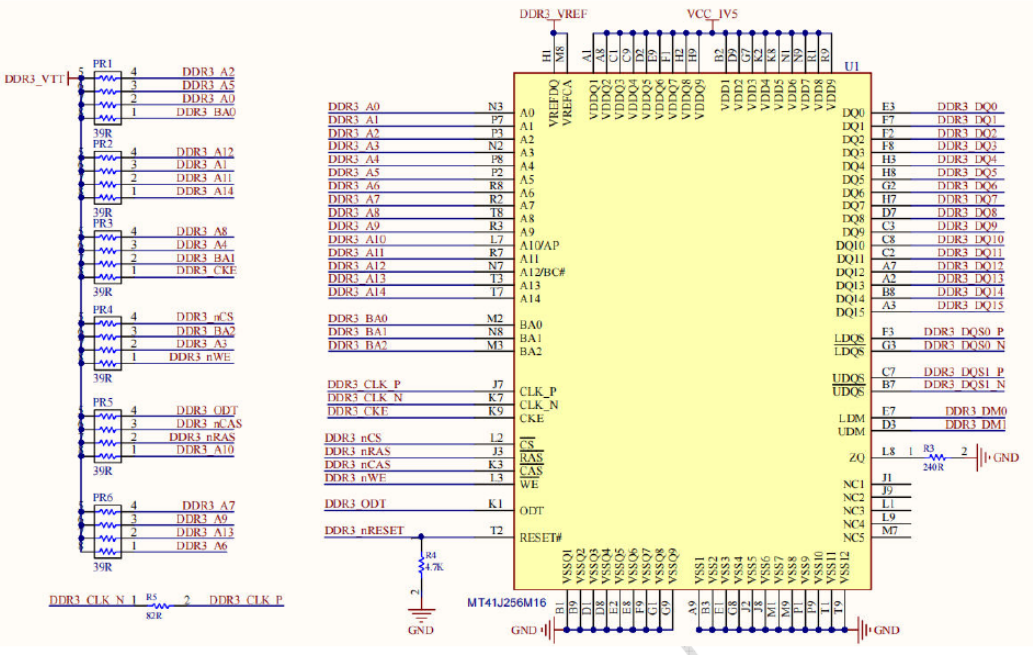
4. Debugger and on-board UART



Schematic part 3.

- **JTAG**: FT232H (USB-to-JTAG).
- **UART**: CH340 USB-to-serial. The CH340 is 3.3 V but PS BANK501 is 1.8 V, so two **74LVC1T45** level translators sit in between. The UART connects to **MIO48/MIO49 = PS UART1**.

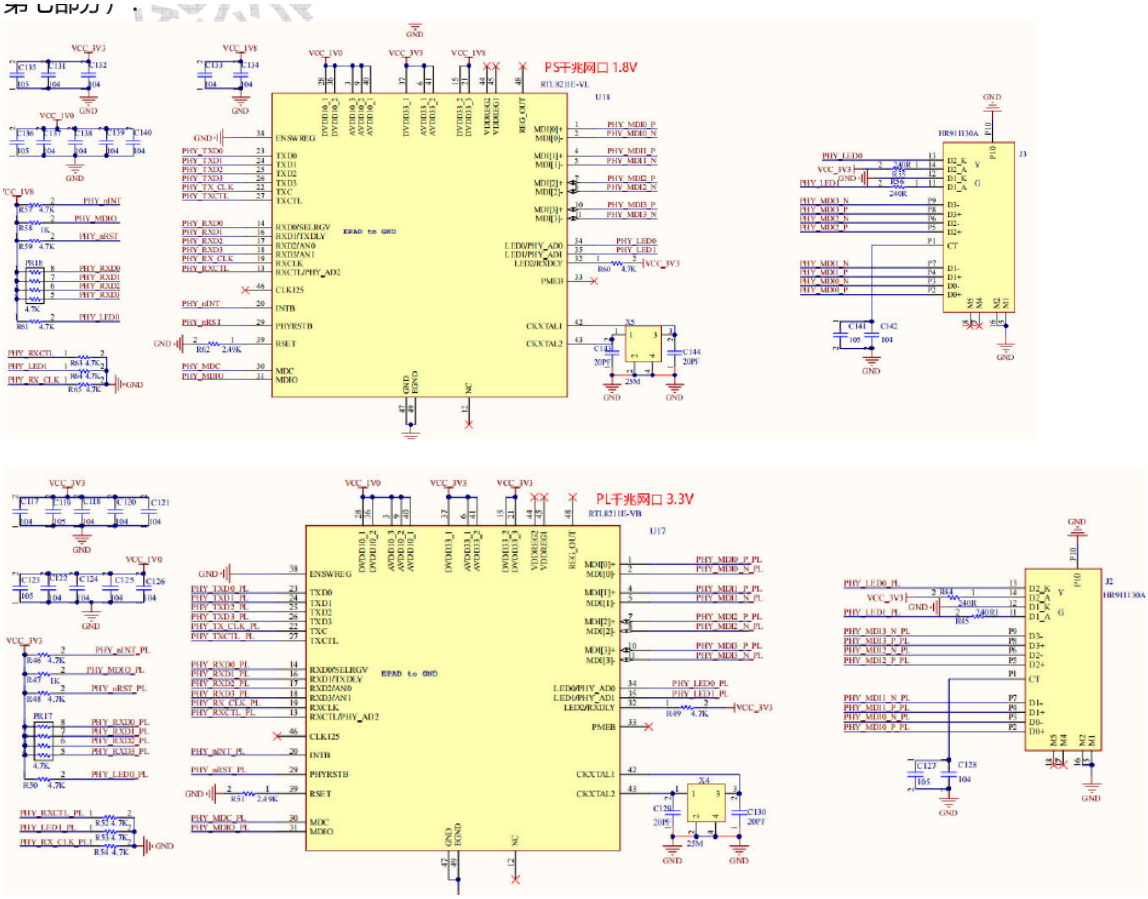
5. DDR3



Schematic part 6. DDR3 is used by the PS only, on BANK502. The board carries a single **MT41J256M16HA-125** (256M × 16 = 4 Gbit = 512 MByte). Address lines are pulled to VTT through 39 Ω resistors (optional for a single chip, kept for stability).

Important: because there is only one 16-bit DDR3 device, the PS DDR configuration must be **16-bit**. The default is 32-bit and will not run unless changed. (Vendor projects use the Vivado part `MT41J256M16 RE-125`, 16 Bit, 533.33 MHz, DDR3_1066F.) [verified]

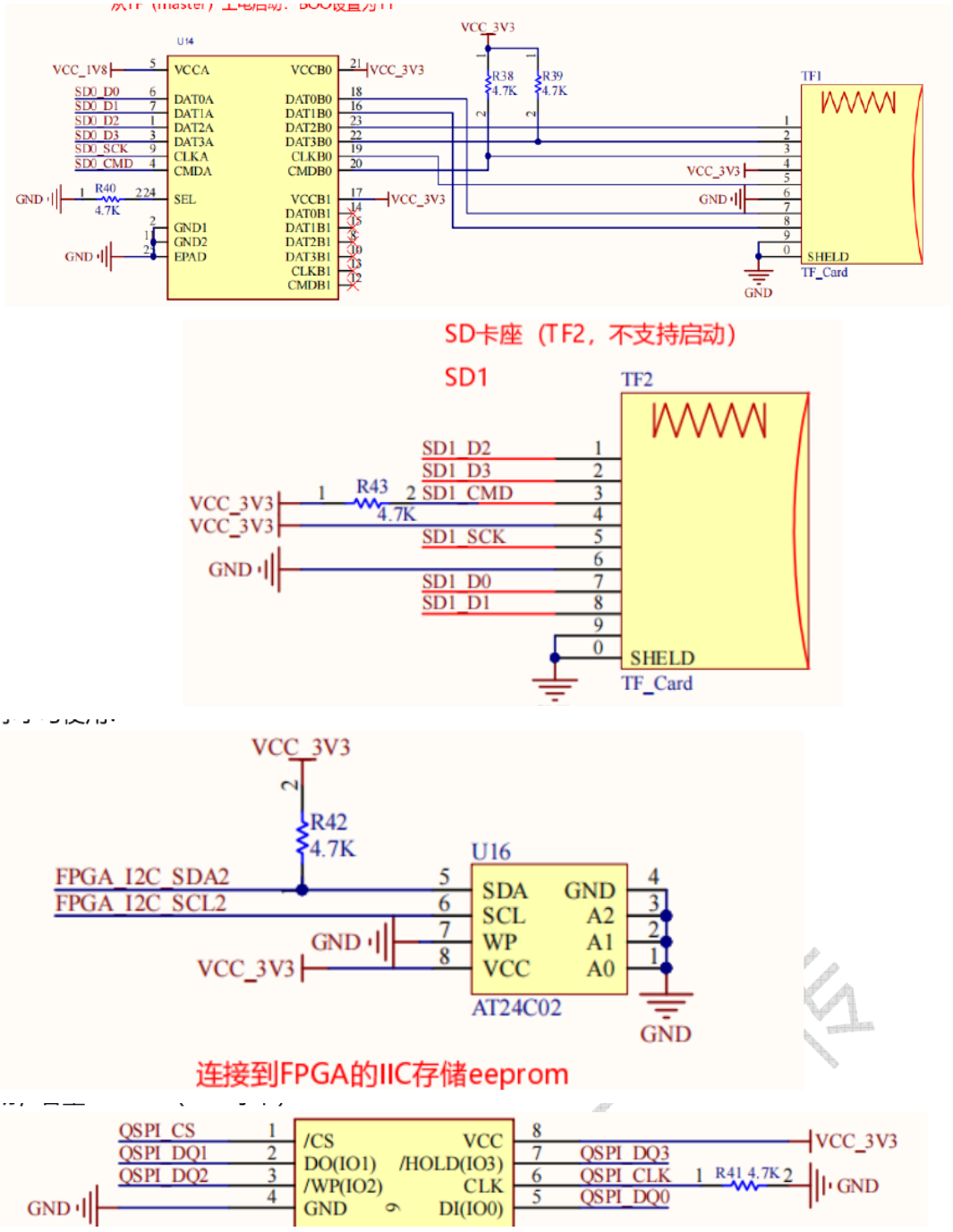
6. Gigabit Ethernet



Schematic part 7. One **RTL8211E-VL** (PS) and one **RTL8211E-VB** (PL); both 1000BASE-T through HR911130A magnetics/RJ45, each with a 25 MHz crystal.

- **PS Ethernet**: RGMII on BANK501 **MIO16–27**, at **1.8 V**; MDIO/MDC on MIO52/53. [verified]
- **PL Ethernet**: on BANK35, at **3.3 V**, RGMII-class pins (see pin table in §12).

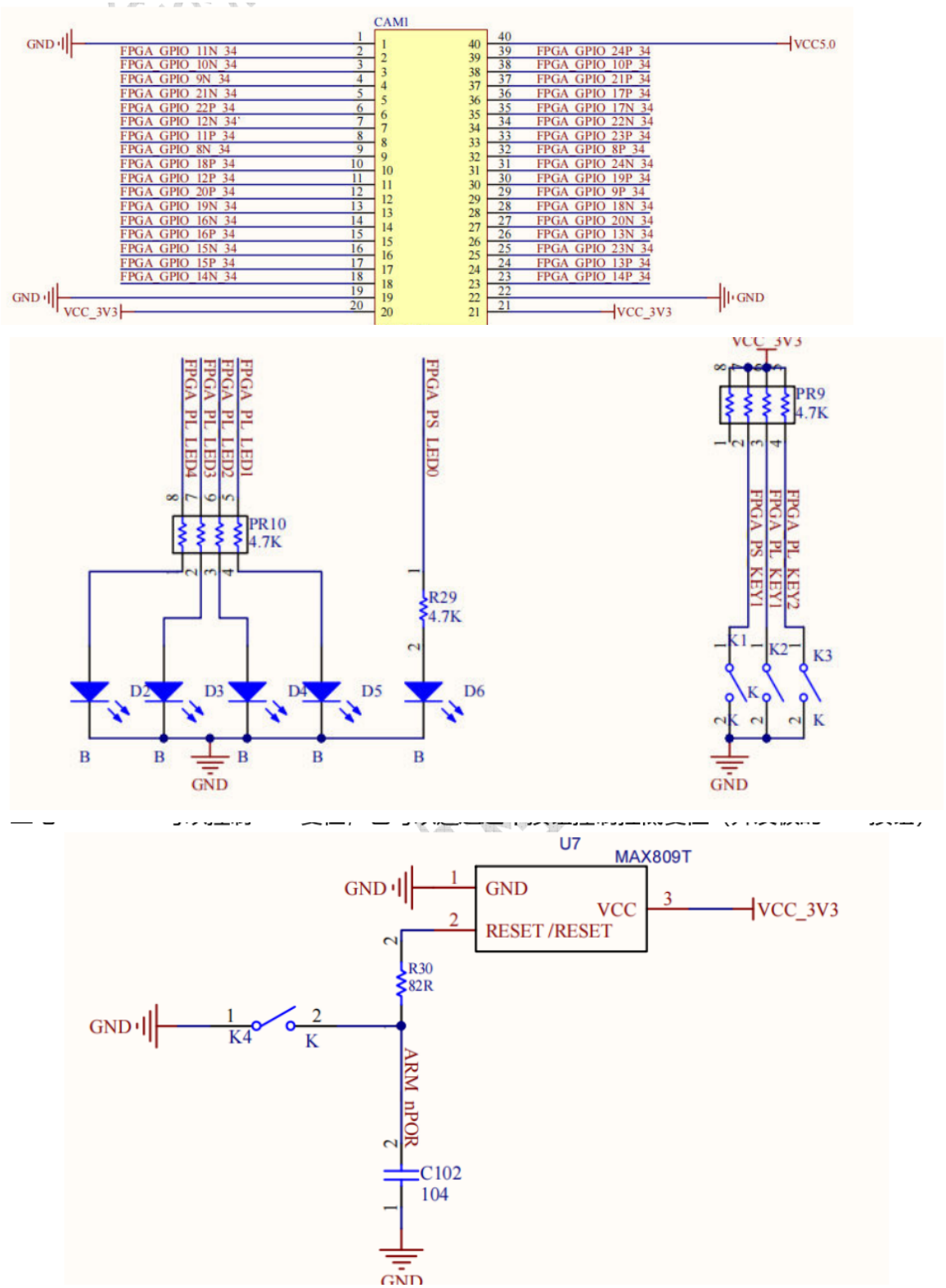
7. Storage



Schematic part 8.

- **DDR3**: see §5.
- **SD0 (TF)**: PS **MIO40–45** on BANK501 (1.8 V). TF cards run at 3.3 V so a **TXS02512**-class SD-specific level translator (the schematic part 8 drawing and the bundled datasheet `TXS02612RTWR.pdf` identify the fitted device as **TXS02612RTWR**, so "TXS02512" in the text is most likely a typo; TF1 is the "master" slot that supports boot, BOOT = 11) is used. No CD/WP detect lines. Supports boot.
- **SD1 (TF2)**: PS **MIO10–15** on BANK500 (3.3 V); no translator required.
- **EEPROM**: 24C02 on the PL (BANK35): **SDA = F20, SCL = F19** (useful for I²C timing practice).
- **QSPI flash**: W25Q128FV, 128 Mbit (16 MByte), PS **MIO1–6** on BANK500; used for QSPI boot.

8. Expansion IO, LEDs and buttons



Schematic part 9.

- **40-pin 2.54 mm header**, 3.3 V, on BANK34. 34 single-ended signals, length-matched to 1615 mil ± 1 mil, suitable for single-ended expansion experiments. Pin assignments are given by the back-side silkscreen (table below).
- **LEDs**: four PL LEDs on BANK34: **LED1 → T12, LED2 → U12, LED3 → V12, LED4 → W13**. One PS LED on **MIO9**. All LEDs are **active high** (signal → 4.7 kΩ → LED → GND). [verified from schematic]
- **Buttons**: PS button **KEY1 → MIO0** (BANK500). PL buttons **KEY2 (PL_KEY1) → M20**, **KEY3 (PL_KEY2) → M19** (BANK35). All buttons pull up to 3.3 V and close to GND, i.e. **active low**. [verified from schematic]

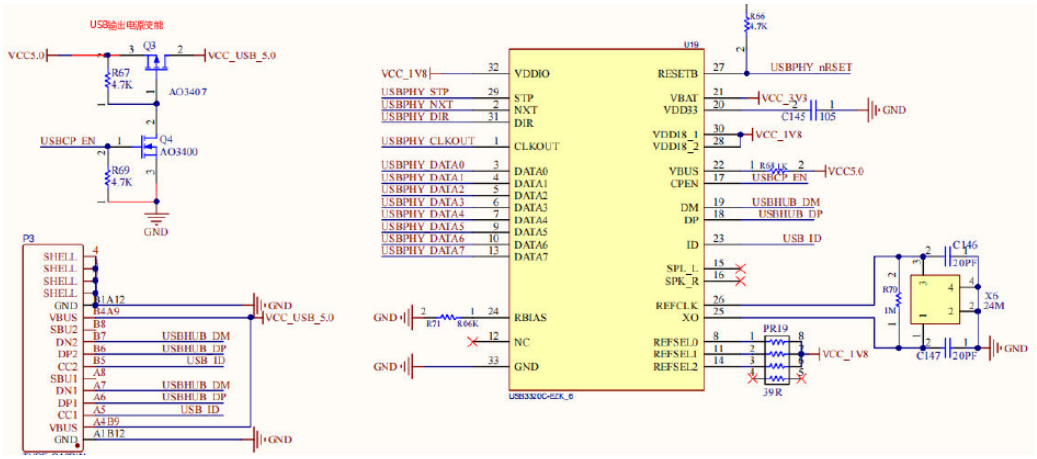
- **POR button** (silkscreen "SYS RST"): schematic part 3, pulls the MAX809T reset line low to reset the PS.

Expansion header pin map (pin numbers follow the schematic's CAM1 symbol: 1–20 down one side, 21–40 down the other; the physical 2×20 header is the silkscreen row/ball table below — from `figures/board_back_silkscreen.jpg`, cross-checked against schematic part 9; right column = header pins 1–20 top to bottom, left column = pins 40–21 top to bottom):

Row	Left (pin)	Left ball	Right (pin)	Right ball
1	40	+5.0 V	1	GND
2	39	P15	2	U15
3	38	V15	3	W15
4	37	V17	4	U17
5	36	Y18	5	V18
6	35	Y19	6	W18
7	34	W19	7	U19
8	33	N17	8	U14
9	32	W14	9	Y14
10	31	P16	10	V16
11	30	R16	11	U18
12	29	T16	12	T17
13	28	W16	13	R17
14	27	R18	14	W20
15	26	P19	15	V20
16	25	P18	16	U20
17	24	N18	17	T20
18	23	N20	18	P20
19	22	GND	19	GND
20	21	+3.3 V	20	+3.3 V

Schematic net names `FPGA_GPIO_nP_34` / `FPGA_GPIO_nN_34` (n = 8…24) correspond to the `IO_L<n>P/N_T*_34` balls (e.g. header pin 2 = `FPGA_GPIO_11N_34` = U15).

9. USB

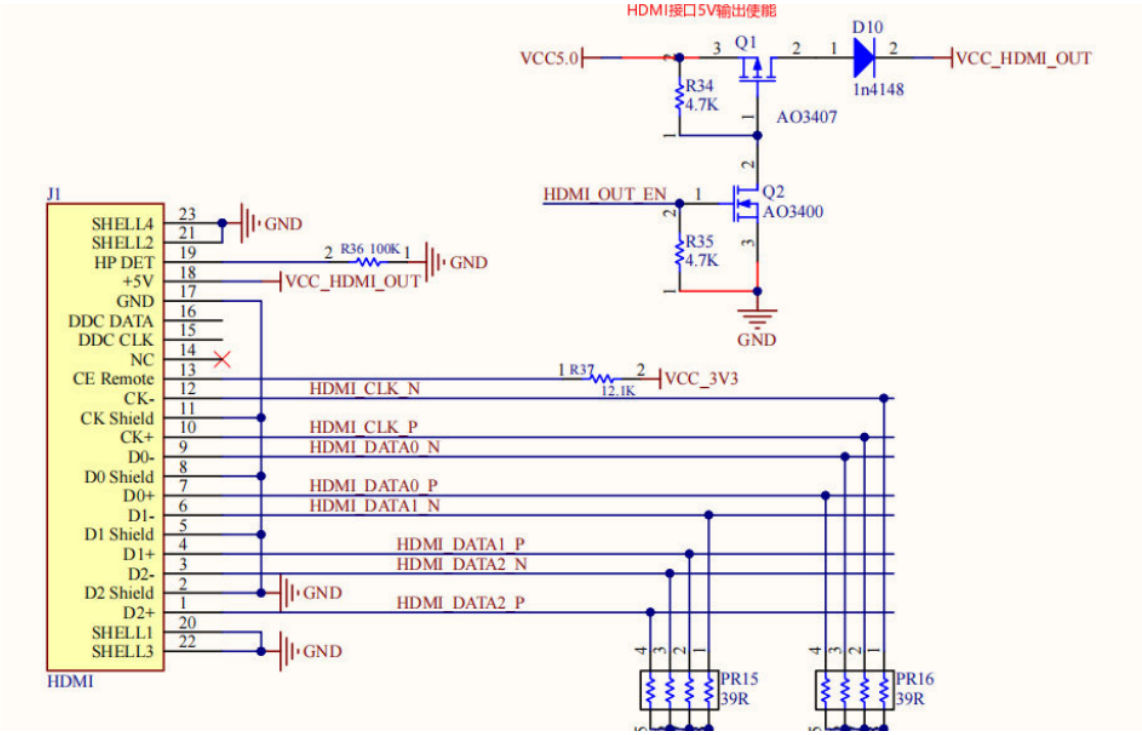


Schematic part 10. This is the USB-C connector in the middle of the three, not the JTAG/UART ones; it is the PS USB 2.0 controller. Older board versions had a hub and supported host only; this version supports **both host and device**. Examples: host role for USB cameras/keyboards/mice under Linux (e.g. PYNQ); device role to appear as a storage device to a PC.

- ULPI PHY **USB3320**, connected to BANK501 **MIO28–39** (1.8 V). PHY reset: **MIO46**. [verified]
- The Type-C port can also **power the board**: the supply gating uses two MOSFETs, and the reverse body diode of the AO3407 lets power flow in from USB (not out). Output power is enabled only when the

USB3320 is running and drives CPEN high — e.g. with the board powered through JTAG/UART, the port can power a USB stick through a Type-C-to-A adapter. For more host ports use an external Type-C hub module.

10. HDMI

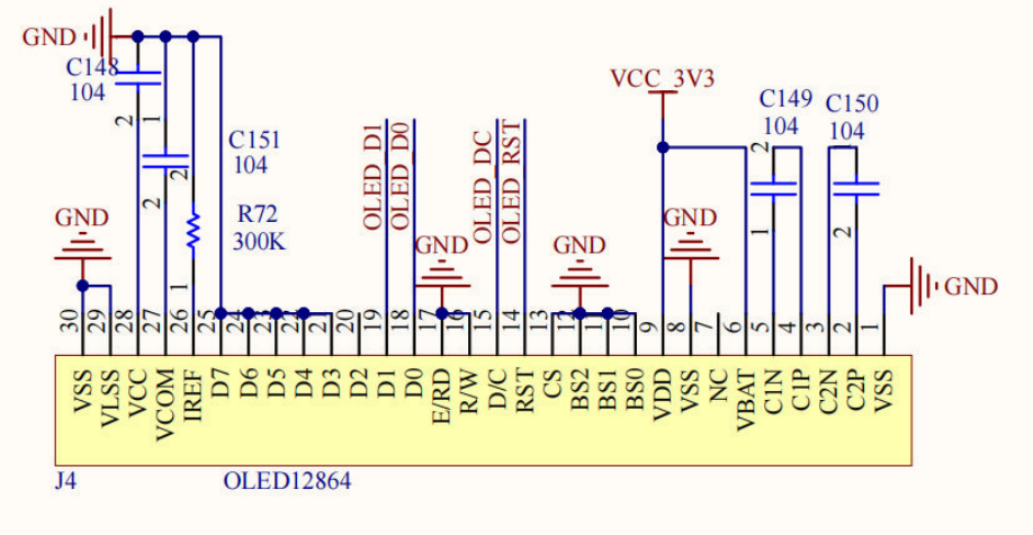


Schematic part 11. **Output only.** The HDMI +5 V is gated by two MOSFETs (controlled by HDMI_OUT_EN); once 5 V is presented, the monitor starts parsing the signal. TMDS pairs go directly to BANK35 pins using the FPGA's 3.3 V TMDS standard; this **requires external pull-ups to 3.3 V of about 50 Ω — 39 Ω is used here**.

Signal	Ball
HDMI_CLK_P / N	H16 / H17
HDMI_DATA0_P / N	D19 / D20
HDMI_DATA1_P / N	C20 / B20
HDMI_DATA2_P / N	B19 / A20
HDMI_OUT_EN	H18

(The vendor `hdmi.xdc` also constrains `HDMI_HPD` to E18, but E18 is OLED_D0 on this board and HDMI is output-only; that line is not valid for this board.)

11. OLED



Schematic part 12. A 0.96" SPI OLED module on PL BANK35 (for simple graphics/text, or for learning OLED timing in the FPGA):

Signal	Ball
OLED_D0 (clock)	E18
OLED_D1 (data)	E19
OLED_DC	F16
OLED_RST	F17

11.1 OLED module notes (from the module's own manual, □□□□□ 0.96" OLED V2.0)

- 128 × 64 pixel panel, driver **SSD1306** with built-in charge pump (8 pages × 128 bytes). Variants: yellow/blue, white, blue.
- The bare panel supports 6800/8080 parallel, 3/4-wire SPI and I²C, selected by BS0–BS2. This board's module is configured as **SPI** (BS0–BS2 = 0; resistors R1, R2, R8 not fitted). For I²C, BS1 = 1, BS0 = 0.
- Module pins: GND, VCC (3–5.5 V), D0 (clock in SPI and I²C), D1 (data in SPI and I²C), RES (active-low reset), DC (data/command), CS (chip select; tied low on this board).
- Porting from the 1.3" SSD1106 panel: that controller has 132 bytes per page, so shift the display address right by 2 or the right 4 columns show wrongly.

12. Pin summary (PL side)

Function	Signal	Ball	Standard	Source
PL clock 50 MHz	PL_CLK_50M	K17	LVCMOS33	spec, XDC
LED1–4		T12, U12, V12, W13	LVCMOS33	spec, XDC
KEY2 / KEY3	PL_KEY1/2	M20 / M19	LVCMOS33	spec, XDC
EEPROM	SDA / SCL	F20 / F19	LVCMOS33	spec, XDC
OLED	D0 D1 DC RST	E18 E19 F16 F17	LVCMOS33	spec, XDC
HDMI	see §10		TMDS_33	spec, XDC
PL Ethernet (RGMII)	RGMII_RESET	G17	LVCMOS33	vendor lwip/Linux XDC
	MDC / MDIO	G18 / G19	LVCMOS33	vendor XDC
	TXC / TX_CTL	J14 / K14	LVCMOS33	vendor XDC
	TXD[0..3]	N16, J19, H20, N15	LVCMOS33	vendor XDC
	RXC / RX_CTL	L16 / L17	LVCMOS33	vendor XDC
	RXD[0..3]	L20, K19, J18, J20	LVCMOS33	vendor XDC
Camera (OV5640 example)	PCLK, HREF, VSYNC, SCL, SDA, RESET	U20, N18, T20, T16, V16, N20	LVCMOS33	vendor XDC

Function	Signal	Ball	Standard	Source
	DATA[0..7]	P18, W16, R17, T17, R18, P19, U18, R16	LVCMOS33	vendor XDC

PS MIO summary:

Peripheral	MIO	Bank/voltage
KEY1	0	500 / 3.3 V
QSPI flash	1–6	500 / 3.3 V
LED D6	9	500 / 3.3 V
SD1	10–15	500 / 3.3 V
PS Ethernet RGMII	16–27	501 / 1.8 V
USB ULPI	28–39	501 / 1.8 V
SD0	40–45	501 / 1.8 V
USB PHY reset	46	501 / 1.8 V
UART1	48–49	501 / 1.8 V
Ethernet MDIO	52–53	501 / 1.8 V

Appendix A. Schematic section titles (translated)

Sheet	Original title	Translation / notes
1	□□□□ ZYNQ□□□□□	Part 1 – ZYNQ power system. Rails: 1.0 V core (MP2145), 1.8 V, 1.5 V, 3.3 V (MP2143DJ), DDR3 VTT/VREF (TPS51206). Power-up order 1.0 V → 1.8 V → 1.5 V → 3.3 V
2	□□□□ ZYNQ□□□□□□□	Part 2 – ZYNQ chip power pins
3	□□□□ ZYNQ□JTAG□□□□□	Part 3 – ZYNQ JTAG and debugger circuit. Power-on-reset supervisor asserts a low reset once 3.3 V is ready; UART and JTAG Type-C connectors can power the board; the PS UART is 1.8 V and is translated to 3.3 V for the CH340
4	□□□□ ZYNQ□PS□□BANK□□□□ARM□	Part 4 – PS banks (ARM only): BANK501 all IO 1.8 V, BANK500 all IO 3.3 V; boot DIP switch; MIO4/MIO5 (QSPI_DQ2/DQ3) double as boot-mode pins
5	□□□□ ZYNQ□PL□□□BANK□FPGA□□□	Part 5 – PL banks; 50 MHz oscillator for the PL
6	□□□□ ZYNQ□PS□□□DDR3	Part 6 – PS DDR3. Layout notes: DQ/DQS/DM length-matched, DQS P/N differential; A0–A14 and control lines length-matched, terminated with 39 Ω to VTT; CLK P/N differential and matched to the address lines
7	□□□□ ZYNQ□PL PS□□□□□□	Part 7 – Gigabit Ethernet for PS (1.8 V, RTL8211E-VL) and PL (3.3 V, RTL8211E-VB)
8	□□□□ ZYNQ□□□QSPI FLASH,SD□□EEPROM	Part 8 – Storage: TF1/SD0 (master, boot capable, BOOT = 11), TF2/SD1 (no boot), W25Q128FV QSPI (boot with BOOT = 10), AT24C02 EEPROM on the FPGA I²C
9	□□□□ □□□□□□□□□□34□IO□□LED□□□	Part 9 – 34 user expansion IOs, LEDs and buttons ("see the silkscreen on the back of the PCB, it names the ZYNQ pins directly")

Sheet	Original title	Translation / notes
10	□□□□ □□□□USB□□	Part 10 – USB circuit. USB output-power enable; when USB is unused or not supplying power as a device it can be a board power input — Q3 has a body diode that conducts
11	□□□□□ HDMI□□	Part 11 – HDMI circuit (HDMI 5 V output enable)
12	□□□□□ PL□OLED□□	Part 12 – PL OLED circuit

Appendix B. Physical design files

- `ZYNQ_MINI□□PCB.PcbDoc` – Altium PCB (binary format 5.01 / "PCB 6.0 Binary File", 23 placed components in the interface-board variant). Components include 0805 passives, the 2.54 mm two-position DIP switch, the 40-pin double-row header, HDMI SMD connector, HR911130A RJ45, OLED096 module socket, TYPE-C connectors and inset TF card sockets. Board sheet 10000 × 8000 mil.
- `ZYNQ_MINI□□□.PcbDoc` – Altium outline for the acrylic cover (no components).
- `ZYNQ_MINI□□□CAD□□□□.DWG` – AutoCAD 2013–2017 (AC1027) drawing for CNC/laser cutting the acrylic cover.
- `ZYNQ_MINI_REVB□□□□.pdf` – single-sheet component-placement silkscreen drawing: shows the connector order along the bottom edge (JTAG, USB, UART), keys K1–K3 and POR on the left, LEDs D0–D4 at the top, OLED096 right, BOOT switch, QFLASH, DDR3, EEPROM, HDMI OUT, SD0/SD1, ETH PS and ETH PL, and EXT IO 3.3 V.

These are proprietary CAD formats; only their metadata and silkscreen were read, not routed copper.

Appendix C. Example-project catalogue (`03□□□□/7010`, Vivado/SDK 2018.3, xc7z010clg400-1)

Archive	Projects
ARM□□ (ARM)	01 UART hello world · 02 MIO in/out · 03 AXI GPIO in/out · 04 AXI GPIO interrupt · 05 RTOS multi-task · 06 SD card FatFs · 07 UART echo · 08 dual-A9 AMP · 09 read/write flash · 10 timer interrupt · 11 PS XADC · 12 FatFs on SD1 (the flip-lid TF slot) · 13 lwIP · 14 TCP flash update
ARM+FPGA□□	01 custom GPIO IP · 02 simple register read/write · 03 custom PWM · 04 HDMI output · 05 AXI PL-write/PS-read · 06 BRAM PL-write/PS-read · 07 XADC · 08 PL OLED · 09 PL lwIP · 10 HDMI SD bitmap · 11 OV5640 camera · 12 dual OV5640
FPGA□□□□	01 LED stream · 02 breathing LED · 03 FPGA flash download · 04 PLL clock · 05 I²C EEPROM · 06 FIFO/ILA debug · 07 single-ended to differential · 08 UART echo · 09 FIFO mark-debug · 10 VGA/HDMI test · 11 RAM read/write · 12 PL read/write of the PS DDR3 · 13 AXI simulation · 14 PL Ethernet MDIO · 15 Ethernet UDP echo
HLS□□	01 LED-stream IP · 02 FIR filter IP · 03 FIR with ModelSim · 04 image-processing intro · 05 floating-point sum/mean co-processor · 06 camera image convolution (Sobel)
LINUX□□	01 PetaLinux 2018.3 boot · 02 Linux with graphical desktop (1920×1080 HDMI) · 03 USB gadget (mass-storage)
MIcroblaze□□□□□	01 MicroBlaze hello world · 02 MicroBlaze UART printf
PYNQ□□□□	01 porting/building PYNQ on Zynq · 02 hello world · 03 GPIO · 04 HLS AXI-Lite multiplier · 05 HLS Sobel · 06 USB-camera face detection

Notes from the vendor `.txt` files (translated):

- ARM: "Projects that use the PS: rebuild the SDK project following the tutorial, otherwise the C code cannot find its paths." Project 12 uses **SD1 (the flip-lid TF slot)**.
- FPGA 12 (DDR3): the PL reads/writes the PS DDR3 over AXI; keep the on-line debug (ILA) signals; the SDK test project must be run with "program FPGA" ticked because clocks and resets exist only once the

processor has started; refresh the device in Vivado afterwards; extract the project near the root of C:.

- FPGA 14: PL Ethernet MDIO example; with a gigabit cable to a PC, two of the four FPGA LEDs light to show a gigabit link.
- FPGA 15: set the PC to 192.168.1.3 and use a UDP debug tool; the example was written for two ports (ETH1/ETH2) and its constraints are in the stale, wrong-device pin block mentioned below.
- HLS 04: create a new HLS project, choose the device for your own board, add the .cpp/.h as sources and tb.cpp as the testbench.
- OV5640 camera: re-map pins if a different vendor's module is used; **do not plug the module into the expansion header backwards** (short-circuit risk); copy the project to a disk root.
- PYNQ 06: needs a UVC (driver-free) 720p USB camera.
- Linux USB-gadget test: mount the SD card, `insmod libcomposite.ko` and `usb_f_mass_storage.ko`, create a 200 MB backing file, configure a `mass_storage.ms0` function under configfs and bind UDC `ci_hdrc.0`. Device trees set `dr_mode = "host"` (desktop example) or `"peripheral"` (gadget) with a ULPI PHY.

Appendix D. Bundled datasheets (□□□□□□.zip)

All are the manufacturers' own English documents and need no translation. Zynq: DS187 (XC7Z010/20), DS190, DS191, UG585 (TRM, 1845 pp.), UG821, UG865 (package pinout), UG933 (PCB guide), UG480 (XADC), PG054, UG476. 7-series: DS180–DS182, UG429, UG470, UG471, UG472, UG474, UG475, UG479, UG586, product guides. Peripherals fitted on the board: MT41J256M16 (DDR3), W25Q128FV, RTL8211E, USB3320C, FT232H, CH340, MAX809T, MP2145, MP2143DJ, TXS02612RTWR, 74LVC1T45, AO3400, 24LC04B-family EEPROM (the schematic shows an AT24C02).

Appendix E. Other files in the vendor package

Path	Content
`01□□□□□□□□/oled□□□□/`	OLED module manual (translated in §11.1), SSD1306 and panel specs, I²C/SPI schematics, demo code, PCtoLCD2002 font-extraction tool and tutorial
`08□□□□□□□□□/*.xdc`	Common constraint files (clock, LED stream, Ethernet loopback, HDMI, OLED, OV5640+HDMI)
`03□□□□/□□PS□□□□□□□□SDK□□□□□c□□□□□□□.txt`	See Appendix C

Caution on the vendor constraint files: `eth_loopback.xdc` and the project `fpga_15_eth_udp_eco` contain a stale block of assignments (T26, AH26, AJ27, AB28, AG29, ...) belonging to a different, larger device; those balls do not exist on the CLG400 package. Only the later block (G17, G18, G19, J14, ...) is valid. Likewise `fpga_05_pl_iic_eeprom` has an older intermediate constraint using G17/G18 for I²C; the final one uses SDA = F20, SCL = F19, which matches the schematic. `clock.xdc` declares `phy_rxc` with an 8 ns period (125 MHz RGMII receive clock at 1 Gbps).

Appendix F. Additional details found in the tutorial sources

Findings from a pass over the hand-written HDL, C, HLS, Python and device-tree sources of the example projects (about 270 files, excluding Xilinx-generated IP/BSP code and multi-megabyte image arrays) and over the 13 tutorial PDFs bundled in the projects. Items are new relative to the spec, schematic and constraint files, unless marked "confirms".

Topic	Finding	Source
DDR3 size	The linker scripts place DDR at `0x00100000` with length `0x1FF00000`; with the first 1 MiB reserved this is the full **512 MiB** window. **Confirms** the 512 MByte printed on the board photo and the 4 Gbit x16 part	`lscript.ld` in the ARM projects
PS user key / LED	`PS_KEY_MIO = 0`, `PS_LED1_MIO = 9`. **Confirms** KEY1 = MIO0 and LED D6 = MIO9	`arm_02_mio_inout helloworld.c`

Topic	Finding	Source
PL PHY MDIO address	The PL RTL8211E answers at **PHY address 1** (`PHY_ADDR = 5'b00001`). PHY-register reads give link/speed (`01` = 10 M, `10` = 100 M, `11` = 1000 M); two of the four LEDs light on a gigabit link	`mdio_driver.v`, `mdio_read_write.v`, `fpga_14` notes
PL RGMII receive path	Uses IDELAY on RXD/RX_CTL with a **200 MHz** IDELAYCTRL reference clock (`REFCLK_FREQUENCY = 200.0`), so a 200 MHz clock must be generated in the design. RX clock is constrained at 8 ns (125 MHz)	`rgmii_rxd.v`, `clock.xdc`
Provenance of the Ethernet example	`rtl8211e_mdio.v` still carries the header "kintex 7 ECO development board" — the example was ported from an older board, which explains the stale Kintex pin block in the XDC files	`rtl8211e_mdio.v`
Default network settings	UDP example: MAC `12:34:56:78:90:ab`, IP `192.168.1.10`, UDP port 1234 (PC set to 192.168.1.3). lwIP examples: IP `192.168.1.5`, MAC `00:0a:35:00:01:02`. TCP flash-update server: port **8080**, files up to **16 MB** (the whole W25Q128 flash)	`udp_top.v`, `udp_txd.v`, lwIP `main.c`, `qspi_remote_update.h`
HDMI limits	Comment in the video-mode table (written for a 7035 variant, where HDMI is on an HR bank limited to 1200 Mbit/s per lane): 1080p60 is not supported and **"this board's HDMI supports at most 1920×1080 @ 50 Hz"**; the TMDS pins here are likewise on HR-bank BANK35, so treat 1080p50 as the practical limit (720p60 and 1080p50/30 are provided). Pixel clock 74.25 MHz, serial clock ×5 (371.25 MHz) in the TMDS encoder	`vga_modes.h`, `hdmi_encoder_top.v`
HDMI hot-plug detect	`hdmi.xdc` and the Linux project constrain `HDMI_HPD` to E18, but E18 is OLED_D0; the schematic ties HP DET to ground through 100 kΩ, so HPD is **not** wired to the FPGA	schematic part 11, `zynq_linux_desktop.xdc`
Dual camera (OV5640) pinout on the expansion header	Camera 0: PCLK U20, VSYNC T20, HREF N18, RESET N20, D[7:0] = R16, U18, P19, R18, T17, R17, W16, P18. Camera 1: PCLK Y14, VSYNC W14, HREF Y18, RESET W18, D[7:0] = P15, V18, V17, V15, U19, W19, Y19, U17. SCCB (I²C) is bit-banged through **PS EMIO GPIO 54–57** (SDA0 = 54, SCL0 = 55, SCL1 = 56, SDA1 = 57); matching the `emio_sccb_tri_io[0..3]` pins in order gives SDA0 = V16, SCL0 = T16, SCL1 = W15, SDA1 = U15 (the V16/T16 pair also matches the single-camera XDC). OV5640 8-bit write address `0x78`, 24 MHz input, 72 MHz pixel clock, RGB565 output. The project sets `CLOCK_DEDICATED_ROUTE FALSE` for the camera pixel clock because it is not on a clock-capable pin	`top_pin.xdc`, `ov5640_reg_conf.h`, `ov5640_init.c`
Header pins used as a UART	The `uart_eco` and MicroBlaze examples use an external USB-UART on the header: **RXD = P15, TXD = U15**. (The on-board USB-UART goes to the PS, not the PL)	`uart_eco.xdc`, `microblaze.xdc`

Topic	Finding	Source
Examples written for a 4-key board	Several C examples print "Press PL KEY1 ~ KEY4" and read `& 0x0F`; this board has only **two** PL keys (M20 and M19) plus the PS key. The XDCs bind a single key (M19) to the GPIO input	`helloworld.c` (arm_03/04/05)
XADC	The PS XADC example reads on-chip temperature and VCCINT, VCCAUX, VBRAM, VCCPINT, VCCPAUX, VCCPDRO — useful for checking the rails, no board-specific wiring needed	`arm_11_ps_xadc`
Linux USB	Device trees set the ULPI PHY at `0xE0002000` with `drv-vbus`; `dr_mode` is `host` for the desktop image and `peripheral` for the mass-storage gadget. **Confirms** USB host/device support	`system-user.dtsi`
Tutorial PDFs	The 13 PDFs inside the projects (AXI reference guide UG1037, AXI VDMA PG020, Video-out PG044, AXI4-Stream infrastructure PG085, Video Timing Controller PG016, rgb2dvi) are generic Xilinx/IP documentation; none mention this board. They need no translation	—

None of these findings changes the board files: they affect usage notes, not pin or PS settings. The HDMI 1080p50 limit and the 200 MHz IDELAYCTRL requirement are the two items most likely to trip up a new design.